BGP and Transport Layer (TCP and UDP)

CS 161 Spring 2026 - Lecture 26

Last time: Local-area network attacks

- If an attacker is on your local-area network, it has a lot of power
  - Can read traffic meant for someone else
  - Can modify traffic between honest users
  - Can disconnect honest users from the network

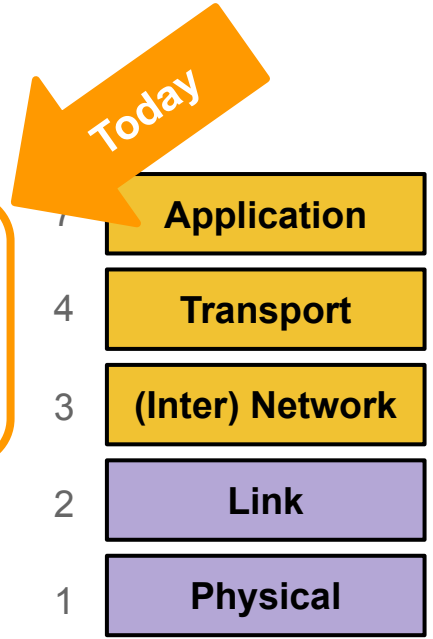
- **Core problem:** There is little authentication/encryption in the network

- **Defense:** Fancier in-network hardware
  - ARP and DHCP attacks: Switches that filter and block traffic that looks suspicious
  - WPA attacks: Enterprise version that has cryptographic authentication

Reminder: Intro to Networking

We can view network protocols in layers:

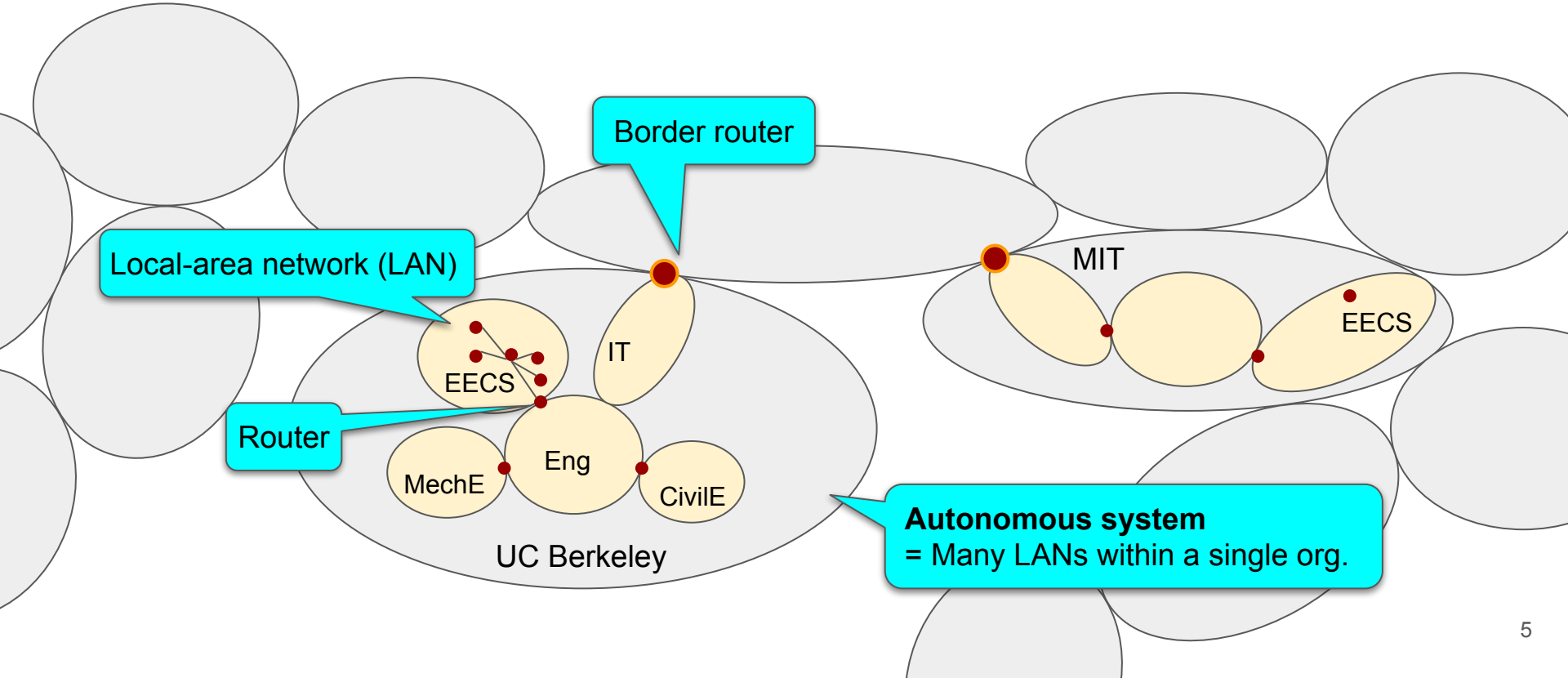
- Layer 7: Applications and services — HTTP, SSH, …
  - Address: Depends on protocol — (e.g., for HTTP: /files/stuff.jpg)
- Layer 4: Transport of data — TCP, UDP
  - Address: 16-bit port number — (e.g., port 80)
- Layer 3: Global packet delivery — IP
  - Address: 32-bit string (IPv4), — (e.g., 10.7.12.130)
- Layer 2: Local frame delivery — Ethernet, Wifi
  - Address: 6-byte string (Ethernet) — (e.g., 4e:05:6c:9a:aa:5a)
- Layer 1: Physical communication — Radio waves, light



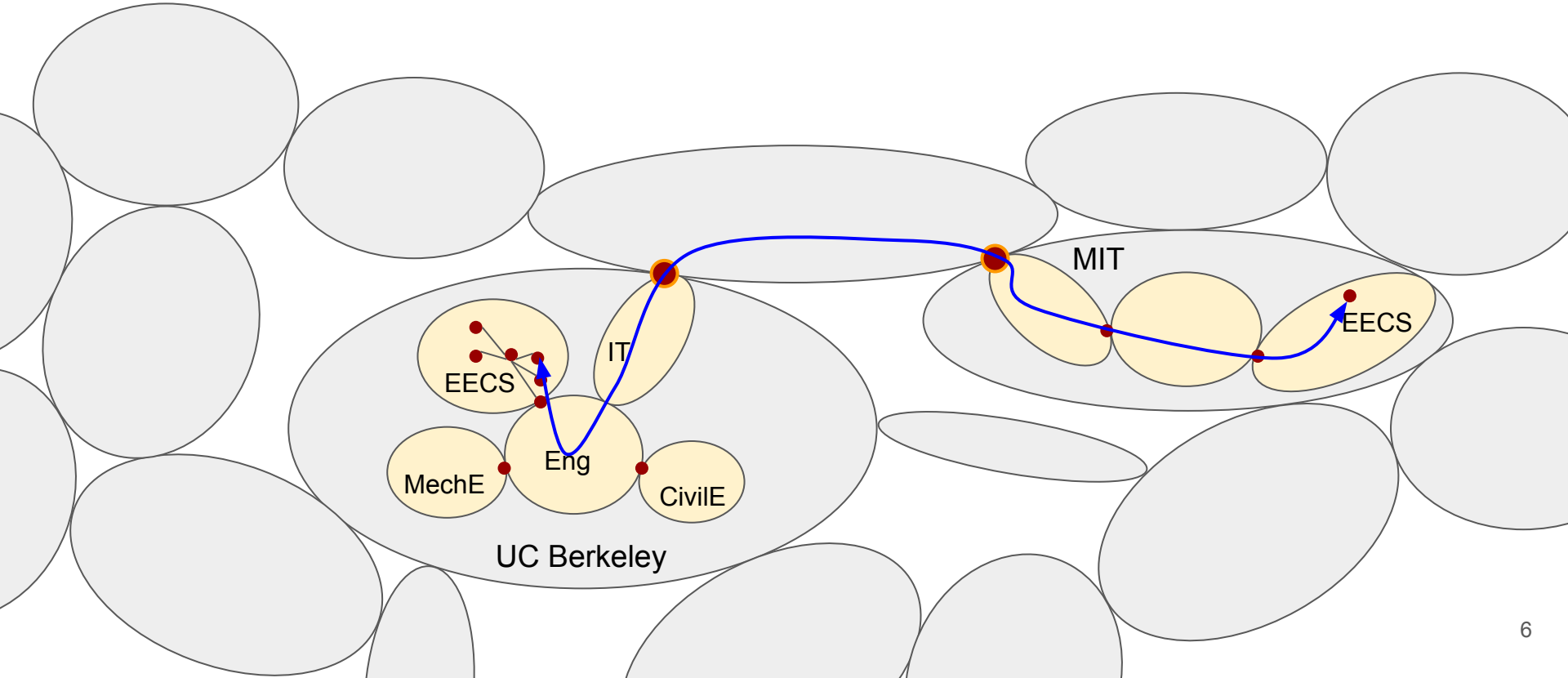
Today: Routing and Internet protocols (TCP and UDP)

- Border Gateway Protocol (BGP) – Routing traffic around Internet
- Transmission Control Protocol (TCP) – Reliably sending data over IP
- User Datagram Protocol (UDP) – Quickly but non-reliably sending data IP

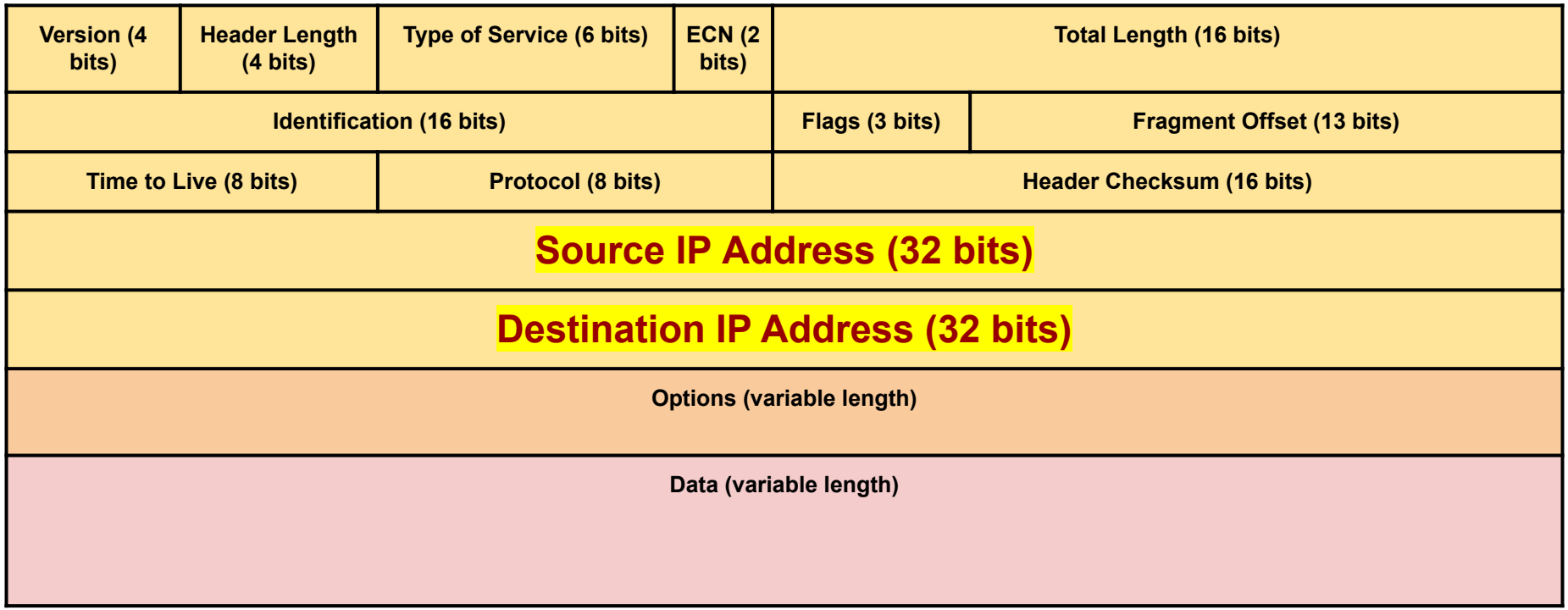
Reminder: Internet topology



Reminder: Distant hosts communicate over IP



IPv4 packet format



Slide courtesy of Grant Ho (UChicago) + my edits

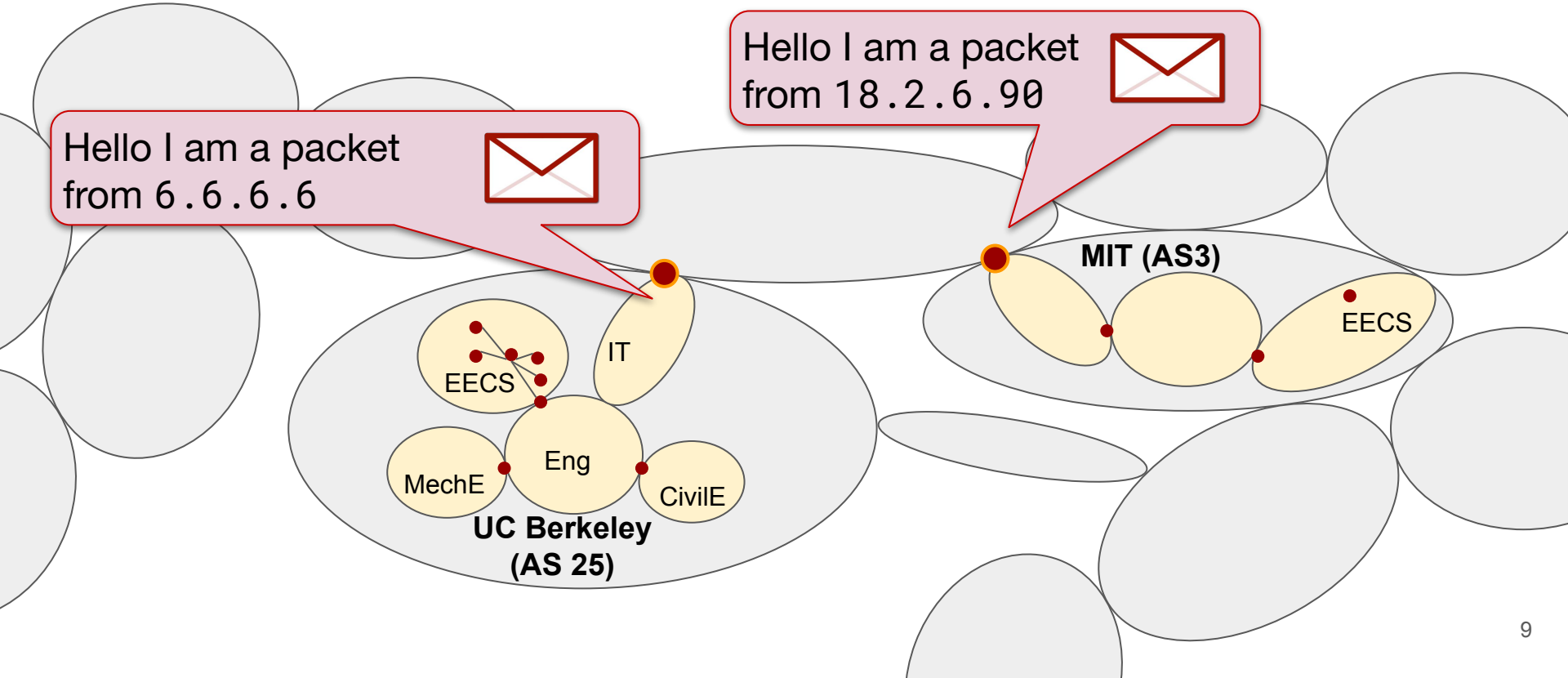
A first attack: IP Spoofing

- Anyone can send IP packets with a fake return/source address
  - Enables packets that look like they're coming from someone else!
  - If you receive an IP packets from 1.2.3.4, it doesn't mean that 1.2.3.4 really sent that packet!
  - IP source addresses are **not** reliable authenticators

- Defense: **Egress filtering** at router
  - Do not let traffic out of your network that pretends to be from outside your network
  - Routers know what source IP ranges sit behind on each of their networks

- Defense: **Ingress filtering** at router
  - Do not let in traffic from outside that pretends to be from inside
  - Berkeley routers will not admit IP packets from the internet pretending to be from Berkeley

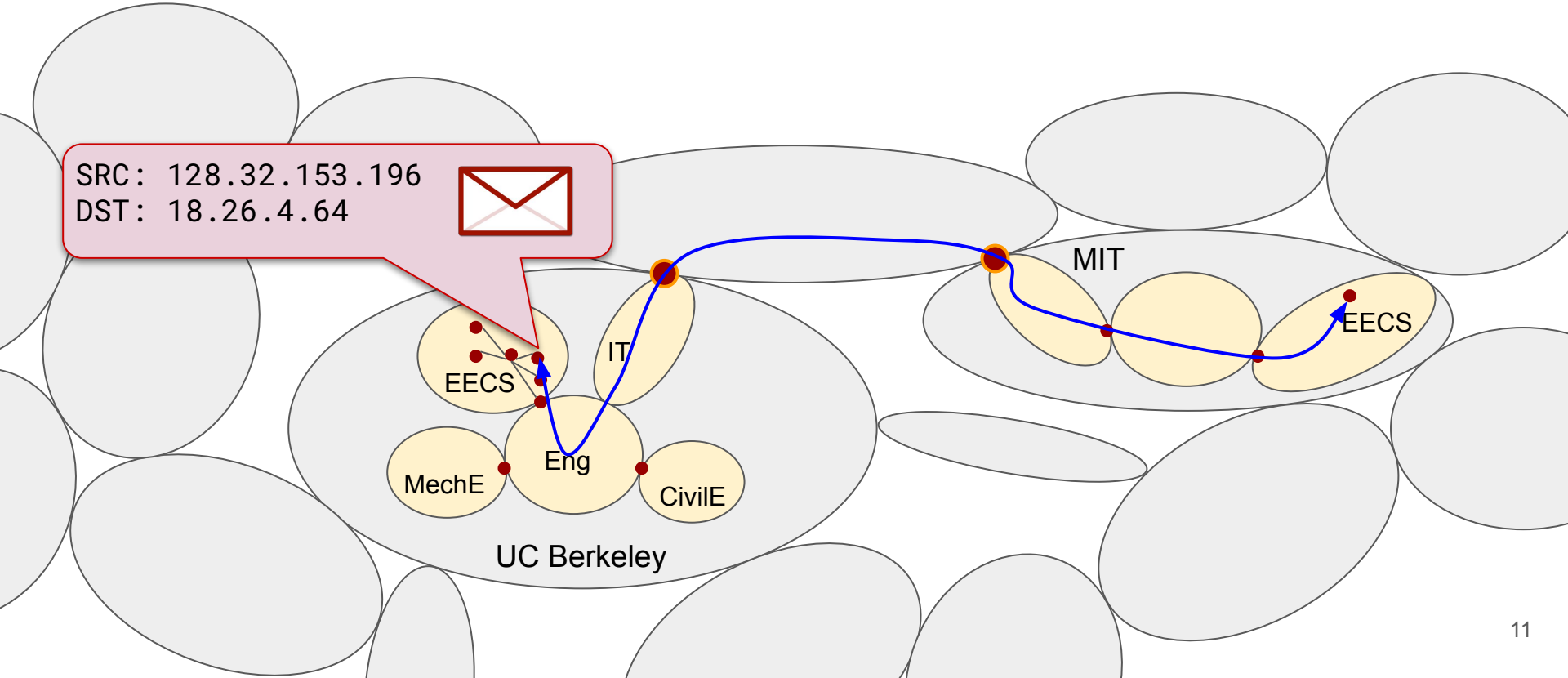
Routers can detect and block spoofed IP packets



Border Gateway Protocol (BGP)

Textbook Chapter 29

Question: How does an IP packet know “where to go”?



IP has a concept of a “subnet”

- Think of subnet = local-area network
  - Doesn’t *need* to be, but usually is
  - You can directly communicate with all machines in your subnet

- How do you determine whether another host is in your subnet?
  - Every IP host is configured with a *subnet mask*
  - Subnet mask looks like an IP address: `255.255.255.0`
  - `subnet = ip_addr XOR subnet_mask`

- Example:
  - Your IP: `192.168. 1. 17 = 0xC0 A8 01 11`
  - Subnet mask: `255.255.255. 0 = 0xFF FF FF 00`
  - Subnet: `192.168. 1. 0 = 0xC0 A8 01 00`

AND

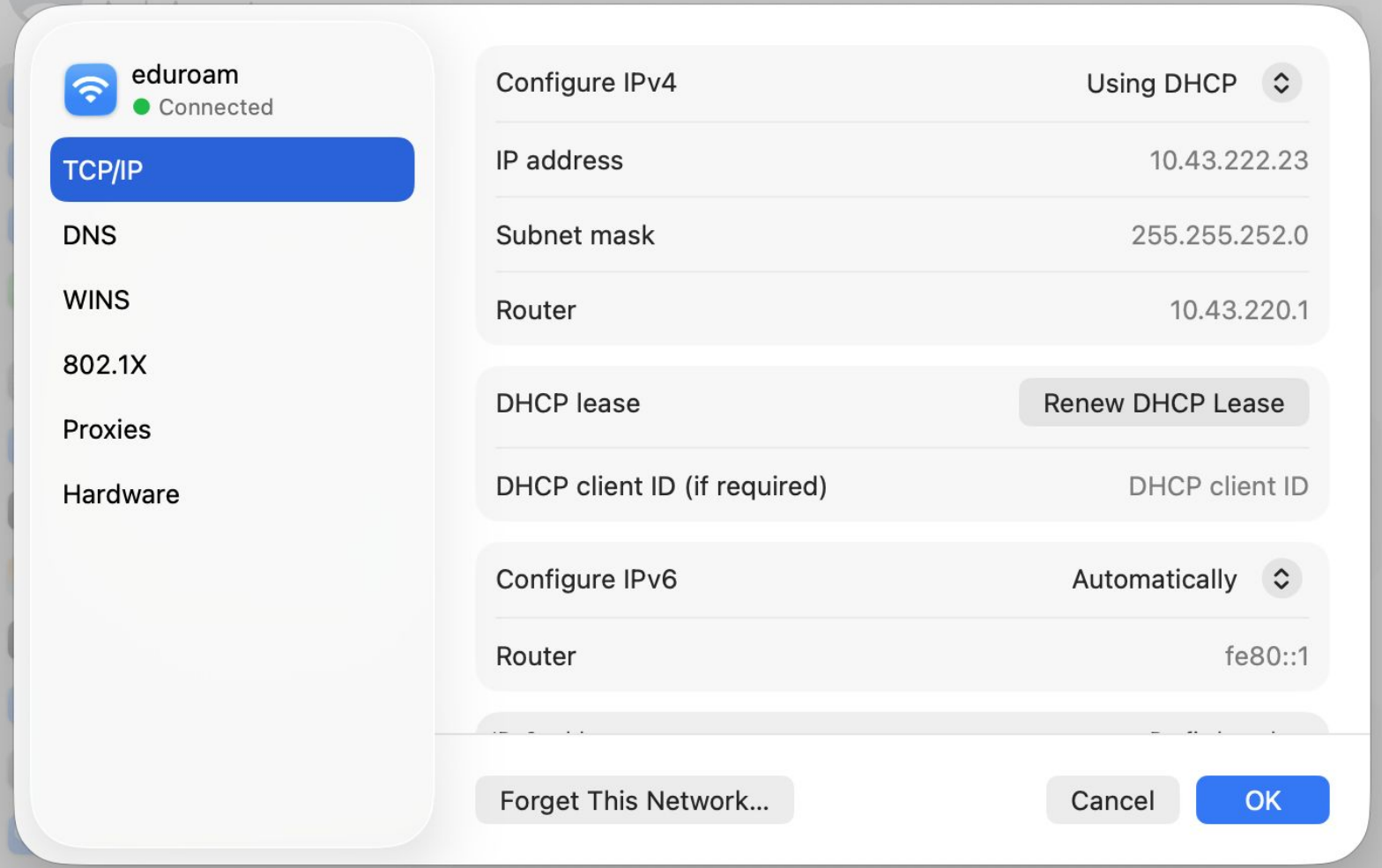
Also written as: `192.168. 1. 0/24` ← “Only the first 24 bits matter”

Your IP configuration

Includes three things:

- IP address: `192.168. 1. 17`
- Subnet mask: `255.255.255. 0`
- Gateway/router: `192.168. 1.24`

The gateway is a special machine inside your subnet.


eduroam
Connected
TCP/IP
DNS
WINS
802.1X
Proxies
Hardware
Configure IPv4
Using DHCP
IP address
10.43.222.23
Subnet mask
255.255.252.0
Router
10.43.220.1
DHCP lease
Renew DHCP Lease
DHCP client ID (if required)
DHCP client ID
Configure IPv6
Automatically
Router
fe80::1
Forget This Network...
Cancel
OK

Question: How does an IP packet know “where to go”?

Answer: For most machines, there is a very simple algorithm

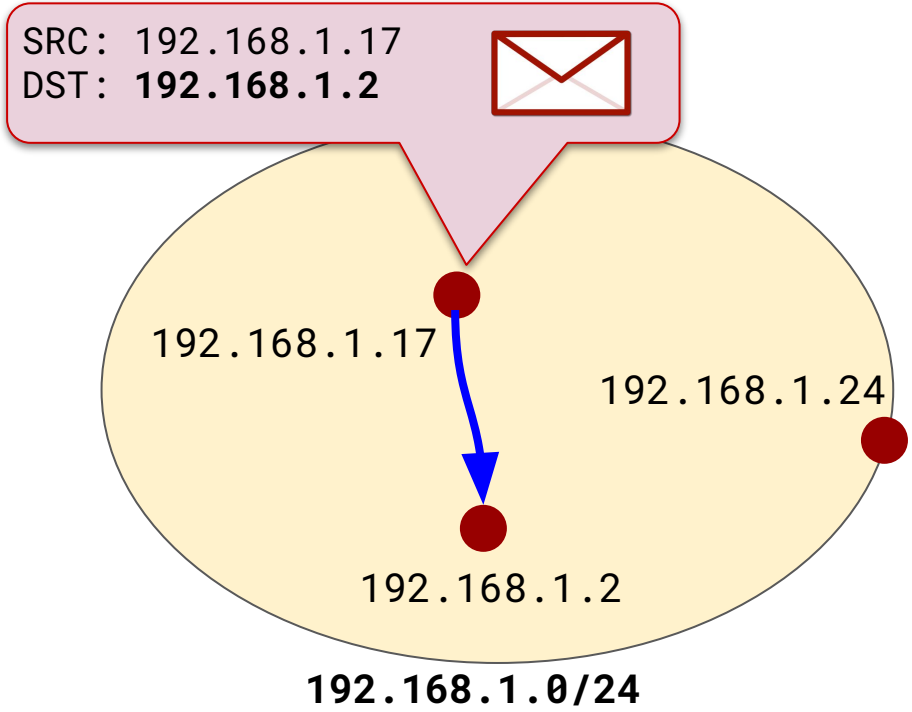
IF destination IP is in my subnet:

- Send packet directly to destination
- E.g., On Ethernet
  - Use ARP to find MAC address of destination IP
  - Send IP packet to destination via Ethernet

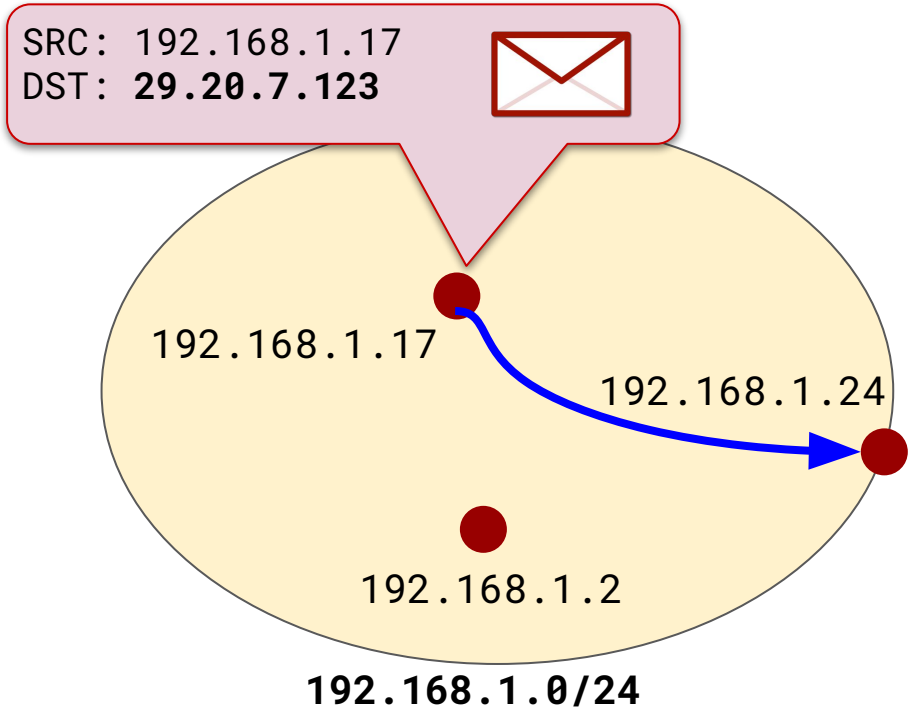
ELSE: // Destination is in some other subnet

- Send packet to gateway (also via Ethernet)

If destination **IS** in subet, send directly to destination



If destination is **NOT** in subnet, send to gateway

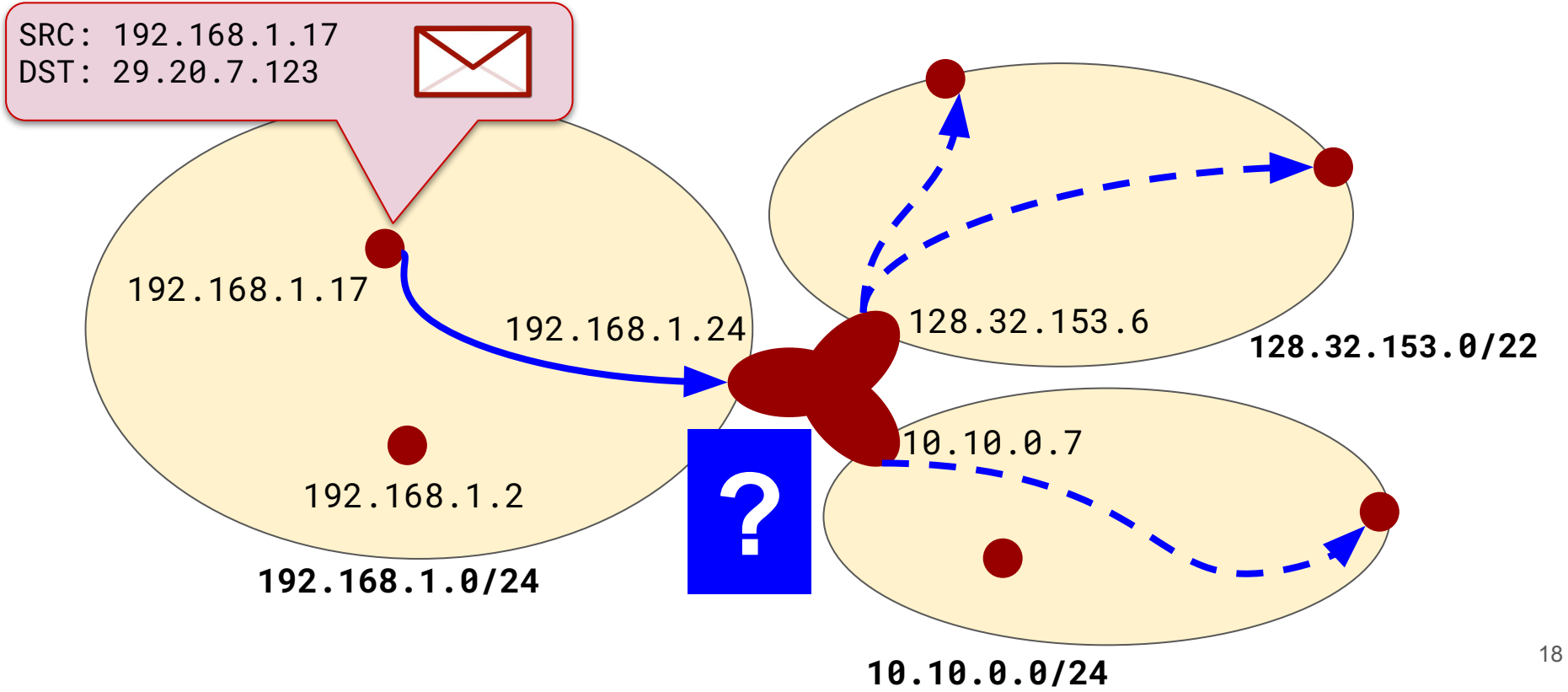


THEN WHAT?

NOT YOUR PROBLEM!!!

THE INTERNET IS GREAT

The gateway router may need to make a hard decision

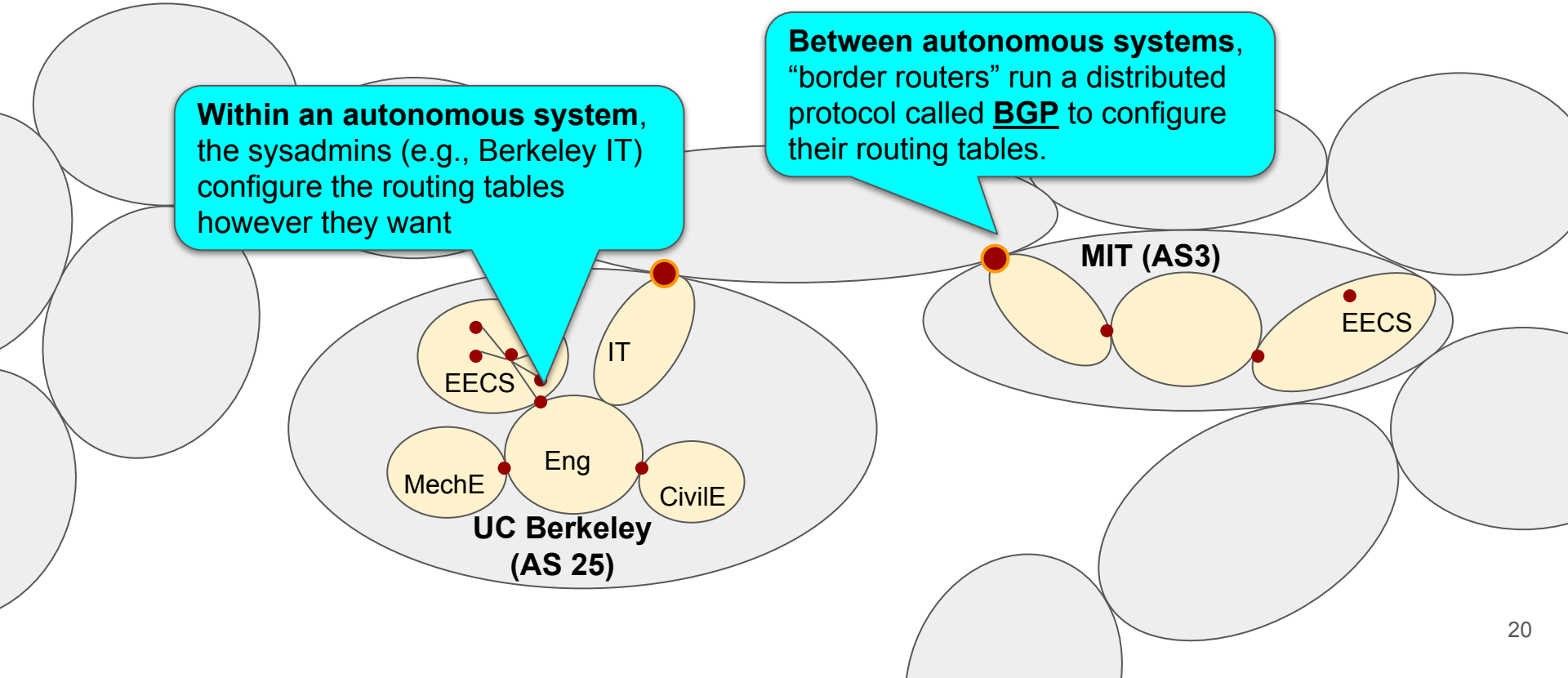


How does router know where to send traffic?

Router maintains a **routing table**

Network	Interface	Gateway/Next Hop
192.168.1.0/24	192.168.1.24	192.168.1.56
192.168.4.0/24	192.168.1.24	192.168.1.56
29.20.0.0/16	10.10.0.7	10.10.0.1
...	...	...
default	128.32.153.6	128.32.153.1

Zooming out: Autonomous systems



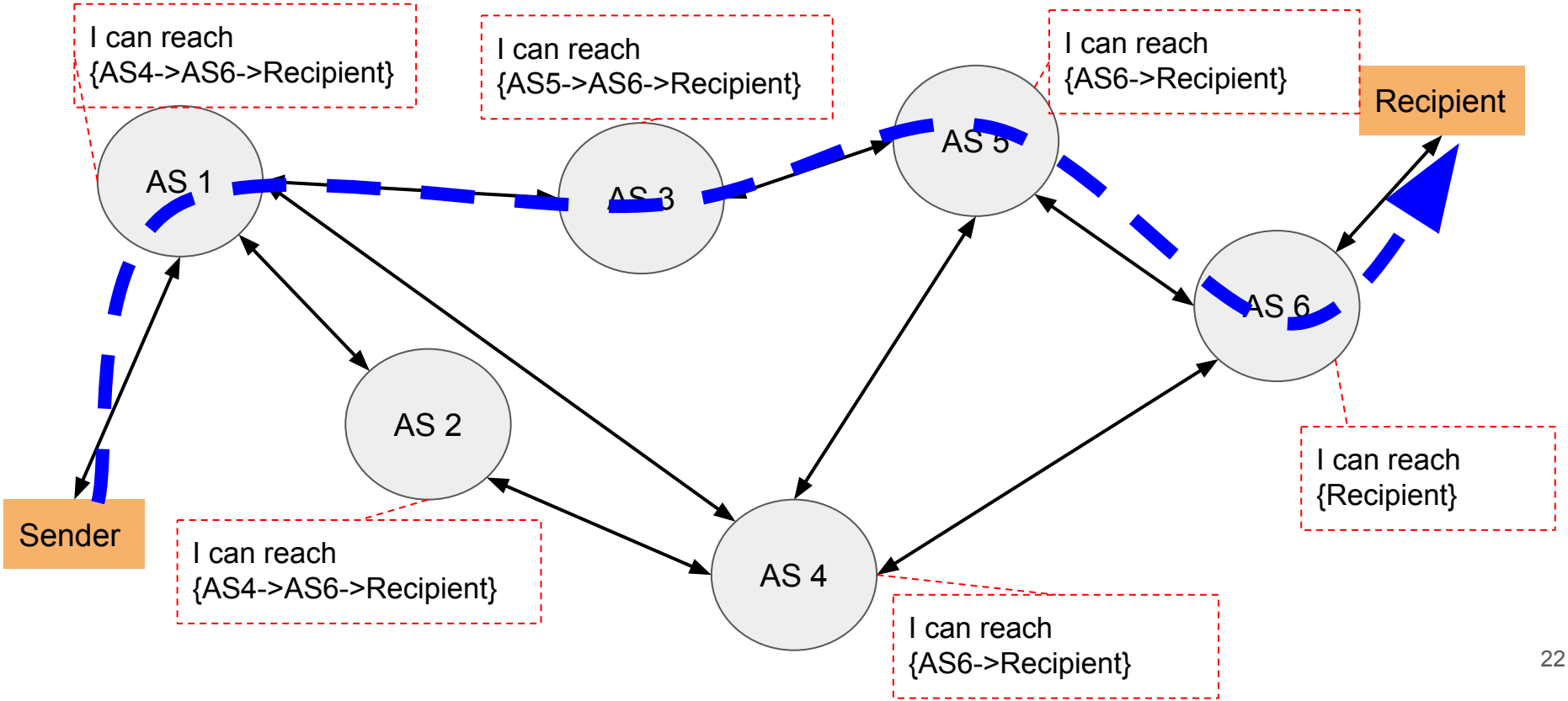
Border gateway protocol (BGP)

- There are 120,000+ autonomous systems (March 2026)
  - Lots of routing information to exchange

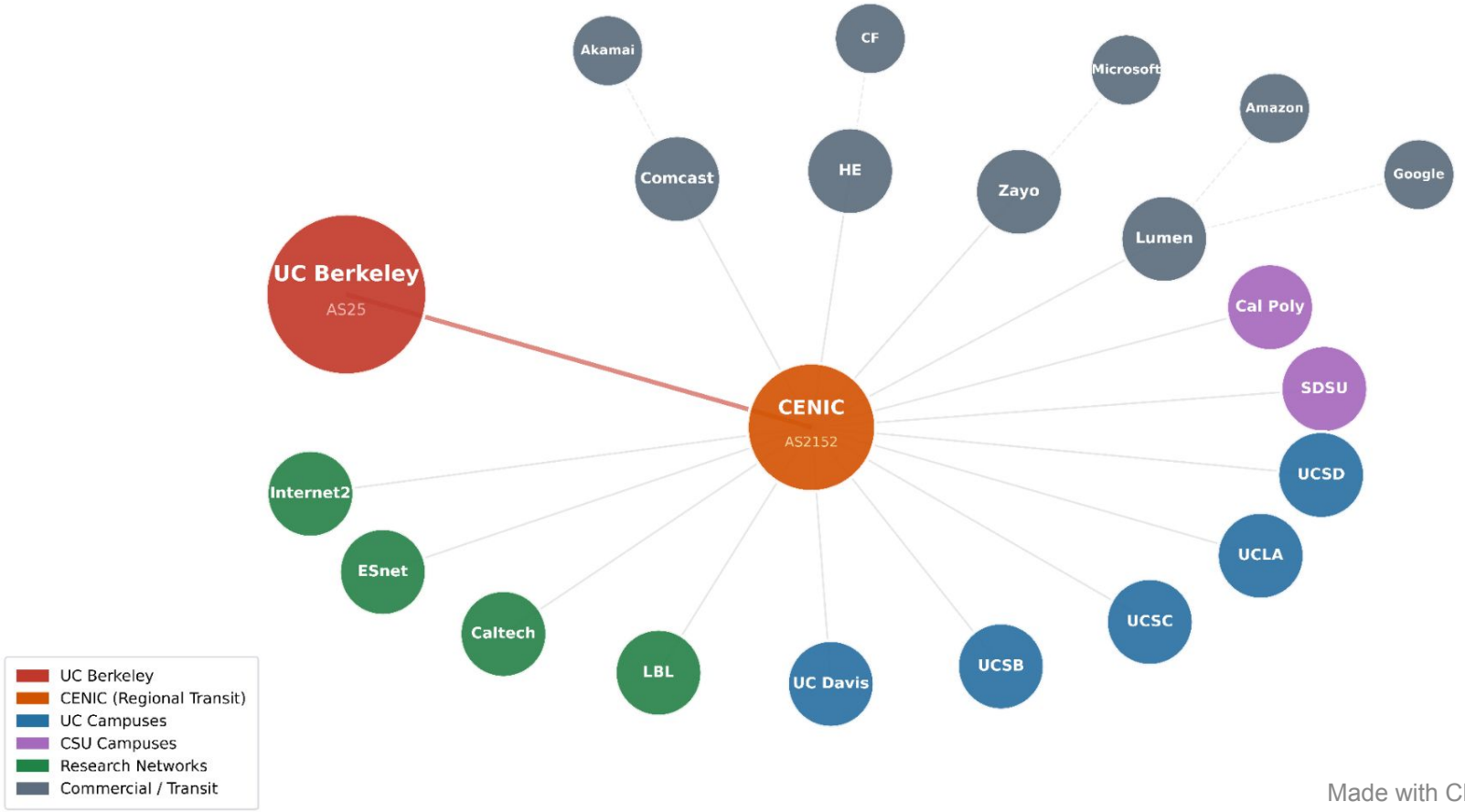
- Operation of BGP:
  - Each BGP router tells its neighbors what IP **subnets/prefixes** it knows how to reach
  - “I can reach `29.20.7.0/24`
    via next hop `192.10.5.13`
    via AS path `4123 → 783 → 714 → 11324`”
  - A length-zero path = “this subnet is in my autonomous system”
  - Internet routers can advertise 1M+ paths

- Border routers use these advertisements to update their routing tables

BGP


I can reach
{AS4->AS6->Recipient}
I can reach
{AS5->AS6->Recipient}
I can reach
{AS6->Recipient}
Recipient
AS 1
AS 3
AS 5
AS 6
AS 2
Sender
I can reach
{Recipient}
I can reach
{AS4->AS6->Recipient}
AS 4
I can reach
{AS6->Recipient}

BGP Peering — AS25 (UC Berkeley) via CENIC



Made with Claude from ripe.net data

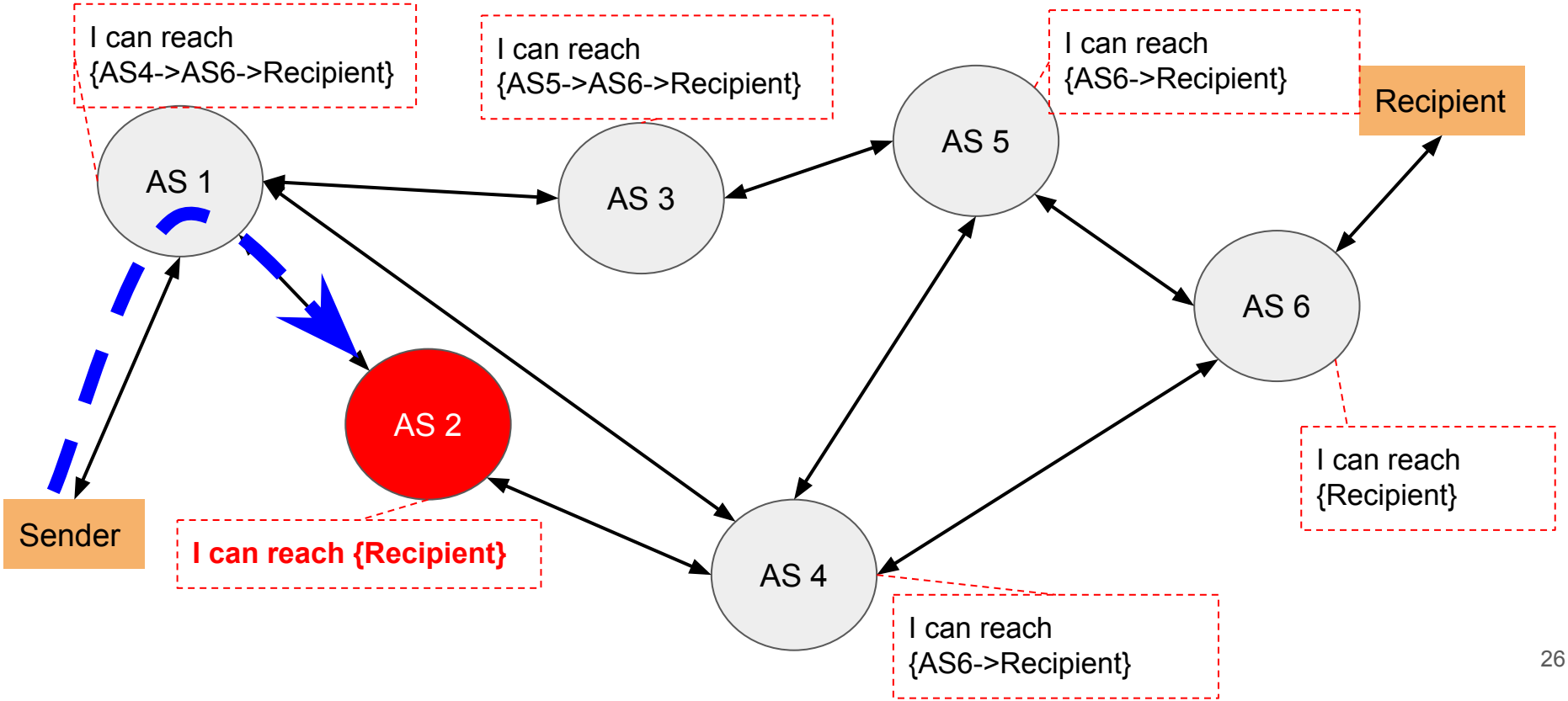
BGP Hijacking

- Border routers can advertise false routes
  - Sometimes because of benign misconfiguration (10.12.4.0/24 looks a lot like 10.12.4.0/2)
  - Or, attacker can make it seem like the shortest path to X is through attacker's network

- E.g., AS3 broadcasts that it is responsible for 128.32.0.0/16 (Berkeley)
  - Example: AS3 broadcasts that it is responsible for 128.32.0.0/16
  - Now, the malicious AS can see and tamper with all traffic for traffic to Berkeley
  - Not easy to detect unless you are looking: Traffic will still get there!



BGP Hijacking



Example: Pakistan government takes out Youtube (2008)

- In 2008, Government of Pakistan wanted to block Youtube domestically
  - Pakistan Telecom advertises “YouTube IPs are inside our AS.”
  - PCCW Global, one of PT’s “peers” (neighboring AS) propagates the route
  - Big chunks of the Internet thought that YouTube was in Pakistan
  - = Black hole for YouTube traffic worldwide for some hours
- Within ~3 hours, PCCW Global withdrew routes announced by PT
- Partial defense: Route filtering
  - Before accepting route, make sure that an AS actually owns IPs it claims it owns
  - … not always easy to check

https://www.ripe.net/about-us/news/youtube-hijacking-a-ripe-ncc-ris-case-study/

TCP and UDP

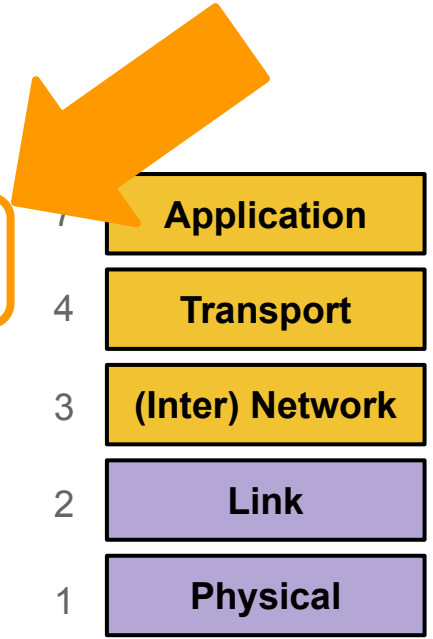


Textbook Chapter 30

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Limitations of IP

- **One IP host only has one IP address**
  - Often one host wants to talk to many servers at once
  - E.g., you have 1000 browser tabs open at once, all talking IP
  - Need some way to figure out which IP packets “belong to” which browser tab
    → UDP and TCP provide **port numbers**

- **IP provides “best-effort” delivery of packets**
  - If you send a packet, it may not arrive
  - Routers may chop up IP packets into smaller packets
  - Packets may arrive in arbitrary order
    → TCP provides **reliable**, **in-order delivery** of a stream of bytes

Ports in TCP and UDP

- **Ports** let us distinguish between different applications on the same machine
  - When **sending traffic**, source port #s can be random
  - When **receiving traffic** (i.e., hosting a public server), use fixed “well known” port numbers

- A few “well-known” ports
  - 22 = SSH server
  - 80 = web server
  - 179 = BGP server
  - 443 = HTTPS server

- Remember: TCP is built on top of IP, so the IP header (and therefore the IP address) is still present